

AI-Based Outfit and Skin Advisor

Complete User Guide

■ Welcome to Your Personal AI Style Assistant

This comprehensive guide will walk you through using the AI-powered outfit and skincare recommendation system. The application uses advanced computer vision and machine learning to analyze your skin tone and type, then provides personalized fashion and beauty advice.

■ Key Features

- AI-powered skin tone detection (Light/Medium/Deep with Cool/Warm undertones)
- Skin type classification (Dry, Oily, Normal, Combination)
- ML-based outfit recommendations for different occasions
- Personalized color palette suggestions
- Skincare product recommendations
- Interactive AI chatbot for beauty advice
- E-commerce integration with shopping links
- Real-time weather integration for seasonal recommendations

■ System Requirements & Setup

Before starting, ensure you have:

- Python 3.8 or higher installed
- All dependencies installed (Flask, OpenCV, scikit-learn, etc.)
- Web browser (Chrome, Firefox, Safari, or Edge)
- Internet connection for weather data
- Camera or image files for skin analysis

■ Step-by-Step Instructions

Step 1: Starting the Application

1.1 Open your terminal or command prompt

1.2 Navigate to the project directory:

```
cd /workspace
```

1.3 Start the Flask application:

```
python app.py
```

1.4 Wait for the server to start. You should see output similar to:

```
* Running on http://127.0.0.1:5000
* Debug mode: on
Starting AI Outfit and Skin Advisor...
Outfit dataset: 48 records
Skincare dataset: 40 records
```

Step 2: Opening the Web Interface

2.1 Open your web browser

2.2 Navigate to the application URL:

```
http://localhost:5000
```

2.3 You should see the AI Style Advisor homepage with:

- Animated gradient background with floating shapes
- Professional navigation header
- Hero section with feature highlights
- Main analysis form on the right side

Step 3: Performing Skin and Style Analysis

3.1 Fill in Personal Information

- Select your Gender: Male, Female, or Other
- Choose the Occasion: Party, College, Office, Marriage, Casual, Formal, Date, or Travel
- Enter your City: This helps determine weather and seasonal recommendations

3.2 Upload Your Photo

- Click the 'Upload Your Photo' area
- Select a clear, well-lit photo of your face
- Supported formats: JPG, PNG (maximum 16MB)
- For testing, you can use the provided test images:
 - /images/SkinTone.jpg (Light warm skin tone)
 - /images/SkinTone.jpg (Medium warm skin tone)
 - /images/SkinToneAnalysis.jpg (Deep warm skin tone)

3.3 Start Analysis

- Click the 'Analyze My Style' button
- Wait for the AI processing (usually 5-10 seconds)
- You'll see a loading animation with 'Analyzing your photo with AI magic...'

Step 4: Understanding Your Results

4.1 Your Profile Summary

The results page displays your uploaded photo alongside:

- Your selected gender and occasion
- Current season and weather data
- Your city and temperature

4.2 AI Skin Analysis

The AI provides detailed skin analysis:

- **Detected Skin Tone:** Light/Medium/Deep + Cool/Warm undertones
- **Detected Skin Type:** Dry, Oily, Normal, or Combination
- **Analysis Process:** Step-by-step breakdown of AI processing

4.3 AI-Recommended Outfit

Personalized outfit suggestions include:

- Specific outfit name tailored to your inputs
- Color palette with 4 complementary colors for your skin tone
- Shopping links to Amazon, Flipkart, and Meesho
- Prediction method badge (ML Model/Dataset Search/Fallback)

4.4 Personalized Skincare Recommendations

Based on your detected skin type:

- Curated product recommendations with images
- Product descriptions and brand information
- Direct links to purchase products
- Custom skincare tips for your skin type

Step 5: Interacting with the AI Chatbot

5.1 Accessing the Chatbot

- Look for the floating chat icon in the bottom-right corner
- Click the circular icon with the chat symbol
- The chatbot window will expand

5.2 Chatbot Capabilities

The AI assistant can help with:

- Skincare routines and product advice
- Outfit coordination and color matching
- Beauty tips and glow enhancement
- Personalized advice based on your detected skin type

5.3 Sample Questions to Ask

- 'What skincare routine is best for my oily skin?'
- 'What colors look good on warm skin tones?'
- 'Give me outfit tips for a party'
- 'How can I make my skin glow naturally?'
- 'What makeup colors suit my skin tone?'

Step 6: Additional Features

6.1 Saving and Sharing Results

- Click 'Save Results' to print or save as PDF
- Click 'Share Analysis' to share your results
- Click 'New Analysis' to start over with a different photo

6.2 Shopping Integration

- Click shopping buttons to browse similar outfits
- Links open in new tabs to preserve your analysis
- Available platforms: Amazon, Flipkart, Meesho

■ Troubleshooting Common Issues

Issue: Application won't start

Solution: Ensure all dependencies are installed with: `pip install -r requirements.txt`

Issue: Image upload fails

Solution: Check file size (max 16MB) and format (JPG/PNG only)

Issue: No skin tone detected

Solution: Use a clear, well-lit photo with visible face. Avoid heavy makeup or filters

Issue: Weather data not loading

Solution: Check internet connection. The app uses fallback data if API fails

Issue: Chatbot not responding

Solution: Refresh the page and try again. Check browser console for errors

■■ Technical Details

AI Technologies Used

- **Computer Vision:** OpenCV for image processing and skin analysis
- **Machine Learning:** RandomForest classifier for outfit predictions
- **Color Analysis:** HSV color space for accurate skin tone detection
- **Texture Analysis:** Local Binary Patterns for skin type classification
- **Weather Integration:** OpenWeatherMap API for real-time data
- **Frontend:** Modern HTML5, CSS3, JavaScript with animations
- **Backend:** Flask web framework with RESTful APIs

Dataset Information

- **Outfit Dataset:** 48 carefully curated outfit combinations
- **Skincare Dataset:** 40+ products across all skin types
- **Categories:** Multiple genders, occasions, seasons, and skin tones
- **E-commerce Links:** Direct integration with major shopping platforms

■ Support & Contact

For technical support or questions about the AI Outfit and Skin Advisor, please refer to the README.md file in the project directory or contact the development team.

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